

CSCI 3230

Homework 3

22.12

For the standard comparison functions, we've looked at a table that showed the number of operations required for each category over various input set sizes; that table is repeated below:

function	n = 10	n = 100	n = 1,000	n = 1,000,000	n = 1,000,000,000
1	1	1	1	1	1
lg n	4	7	10	20	30
n	10	100	1,000	1,000,000	1,000,000,000
n lg n	40	700	10,000	20x10 ⁶	30,000,000,000
n ²	100	10,000	1,000,000	10 ¹²	10 ¹⁸
n ³	1,000	1,000,000	10 ⁹	10 ¹⁸	10 ²⁷
2 ⁿ	1,000	1.3x10 ³⁰			
n!	3.6x10 ⁶	9.3x10 ¹⁵⁷			

Now consider some other questions about the speed of algorithms.

1. Complete the table above.

2. For each function category and time value given below, find the largest input size that can be computed on a machine that runs 1 instruction per microsecond, in the allotted time – for example, how much input can be processed by an n² algorithm in 1 minute?

function	1 second	2 minute	2 hours	2 days	2 months	2 years	1000 years
1	—	—	—	—	—	—	—
lg n	2 ¹⁰⁶						
n	1,000,000	120,000,000	7.2 x 10 ⁹	1.7 x 10 ¹¹	5.1 x 10 ¹²	1.2 x 10 ¹⁴	6.2 x 10 ¹⁶
n lg n							
n ²	1000	120,000	7,200,000	172,800,000	5.1 x 10 ⁹	6.2 x 10 ¹⁰	3.1 x 10 ¹³
n ³	100	12,000	720,000	17,280,000	518,400,000	6.2 x 10 ⁹	3.1 x 10 ¹²
2 ⁿ	19	2280	136,800	3,283,200	99,496,000	1.1 x 10 ⁹	5.9 x 10 ¹¹
n!	9	1080	64,800	1,555,200	46,656,000	559,872,000	2.7 x 10¹¹

3. Order the function by Θ class from lowest to highest.

- 2 F₁(n) = lg n
- 8 F₂(n) = lg n * 3ⁿ
- 4 F₃(n) = n
- 3 F₄(n) = √n
- 9 F₅(n) = n * 3ⁿ
- 7 F₆(n) = 3ⁿ
- 6 F₇(n) = n^{2.5}
- 5 F₈(n) = n²(lg n)
- 10 F₉(n) = n!
- 1 F₁₀(n) = 7